Pablo Diez

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Education

University of Deusto: B.S. in Computer Engineering — Bilbao, Spain
Grade: 8.27/10 | Awarded Honors Distinction in Programming and Software Design — 2023 - 2027

Experience

CAF Signalling — Bilbao, Spain
Incoming Software Engineer Intern — Critical Software Department — October 2026 - May 2027

- Joining the ATS team to develop a **C++ gateway** that bridges train diagnostic data from the proprietary **CDU protocol** to the **NAOS JMS messaging bus**, enabling realtime consumption by ground systems.

CAF Signalling — Bilbao, Spain
Software Engineer Intern — CBCT Testing Department — December 2025 - August 2026

- Engineered the cross-platform migration of large-scale **C++ railway simulation software** from Windows (**MSVC**) to Linux (**g++**) by refactoring **Qt** architecture, migrating from **WinPcap** to **libpcap**, and replacing **windows.h** API functions.
- Automated the deployment pipeline by writing **CMake** configurations and scripting the generation of a **portable container image** with externalized configurations, achieving zero-dependency deployments and eliminating manual setup.
- Validated real-time system integrity and accurate message processing using **Python** scripts and log analysis, and authored comprehensive technical documentation to ensure maintainability.

Skills

- **Skills:** C++, C, Python, Java, SQL | Git, Docker, Linux
- **Languages:** English (Cambridge C1 Advanced, 198/210), Spanish (native), Basque (advanced)

Awards & Extracurriculars

- **Ranked 2nd** nationally on **CTFtime**, cybersecurity competitions, for 6+ consecutive months. Including a **1st place** finish at the *Ikerlan Qualifier* (43 teams) and qualification for the *Midnight Flag* on-site finals in France (364 teams).
- Authored technical articles on **modern C++**: virtual tables, concurrency, memory layout, language internals, synchronization primitives and type traits. Alongside **writeups** for complex CTF challenges, published on my personal website: pablodiez.dev.
- Cofounded the University Engineering Club: 0xdecode.es. Organized and delivered talks and workshops.

Projects

Performance & Compiler Research — **Perf, C, C++**

- Conducted an in-depth analysis of GCC optimization levels (from -O0 to -Ofast) to evaluate their impact on execution speed, memory usage and instruction generation in C and C++ applications.
- Profiled execution metrics using perf and inspected x86 assembly to study low-level compiler behavior, investigating concepts such as use of registers, loop unswitching, tail-call optimizations and function inlining.
- Quantified optimization gains across distinct scenarios: achieved a 79% time reduction (9.48s to 1.98s) in sequential execution. And a 95% drop in function calls (866M to 42M) for recursive approaches. Validating these findings in complex software by recompiling the **gzip** utility from source (-O0 vs. -O2), accelerating directory compression by 35% (54s to 35s).

Fluid Simulation — **C++, OpenGL**

- Developed an interactive real time fluid simulation using the Smoothed Particle Hydrodynamics (SPH) method, utilizing GPU acceleration through OpenGL for efficient rendering.
- Incorporated complex physics models to realistically simulate fluid dynamics like pressure, viscosity and density.
- Optimized performance using spatial hashing techniques to accelerate neighbor searches and reduce computational complexity. Achieving real time frame rates (>60 FPS) with 800+ particles, demonstrating scalability and efficiency.

Tumor Detection in MRI Images — **Python**

- Designed and trained a convolutional neural network (CNN) to identify tumors in real MRI scans using TensorFlow and Keras.
- Utilized AutoKeras to automatically search for the most efficient architecture.
- Trained the model on a dataset of 3,800 MRI images (75% training, 25% validation), achieving an accuracy of 87.83%.